

Taravat Anvari

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Research Interests

Virtual Reality; Redirected Walking; Gaze Behaviour and Visuomotor Adaptation; 3D Pose Estimation; Eye–Body Coordination; Gaze and Motion Analysis; Machine Learning.

Education

PhD in Psychology (Cognitive Neuroscience) 2023–present
University of Münster, Germany

Advisor: Prof. Dr. Markus Lappe

MSc in Application Software 2020–2023
Chung-Ang University, Seoul, South Korea GPA: 4.17

BSc in Computer Science and Information Technology 2014–2018
Sadjad University of Technology, Iran GPA: 3.60

Publications

- Hypki, N.*, **Anvari, T.***, Steinicke, F., Lappe, M. (2026). *Long-term Adaptation in VR: Retention of Altered Sensorimotor Contingencies through Redirected Walking*. IEEE Transactions on Visualization and Computer Graphics.
- **Anvari, T.**, Lappe, M., de Lussanet, M. (2025). *Where Does Gaze Lead? Integrating Gaze and Motion for Enhanced 3D Pose Estimation*. IEEE VR Workshops (VRW).
- **Anvari, T.**, Park, K. (2023). *GeoPose: Geometry-Incorporated Posing of a Full-Body Avatar from Sparse Trackers*. IEEE Access.
- **Anvari, T.**, Park, K. (2023). *Upper Body Pose Estimation Using Deep Learning for a Virtual Reality Avatar*. Applied Sciences.
- **Anvari, T.**, Park, K. (2022). *3D Human Body Pose Estimation in Virtual Reality: A Survey*. International Conference on Information and Communication Technology Convergence (ICTC).

*Equal contribution.

Ongoing and Past PhD Research

Gaze Adaptation Under Redirected Walking 2023–present

Conducting a multi-session redirected walking study with high-frequency binocular eye tracking. Investigating whether gaze behaviour adapts across short- and long-term exposure to curvature gains. Analyzing gaze direction, gaze–head alignment, fixation behaviour, and gaze stability under manipulated and natural locomotion.

Gaze-Enhanced Full-Body Pose Estimation 2023–2024

Collected a synchronized dataset of body motion (120 Hz) and Tobii Eye Tracker (50 Hz) using a motion capture system. Developed a multimodal deep learning model (PoseFusionNet) that integrates gaze with sparse VR tracking (HMD and controllers). Demonstrated improvements in upper-body and full-body joint rotation estimation.

Teaching & Mentoring

Instructor, Bachelor Seminar: Attention Spring 2025, Autumn 2025, Autumn 2026
University of Münster

Teaching Assistant, Chung-Ang University 2020

Technical Skills

Programming: Python, PyTorch, TensorFlow, C++, C#, Unity3D
VR/Tools: SteamVR, SRanipal, MotionBuilder

Data: Eye tracking, motion capture, time-series analysis

Other: Git, AMASS pipeline, MATLAB

Languages

Persian (native), English (C1), Korean (B1), German (A2)

Awards & Scholarships

CAYSS Scholarship, Chung-Ang University

2020

Academic Fellowship, Ferdowsi University

2018